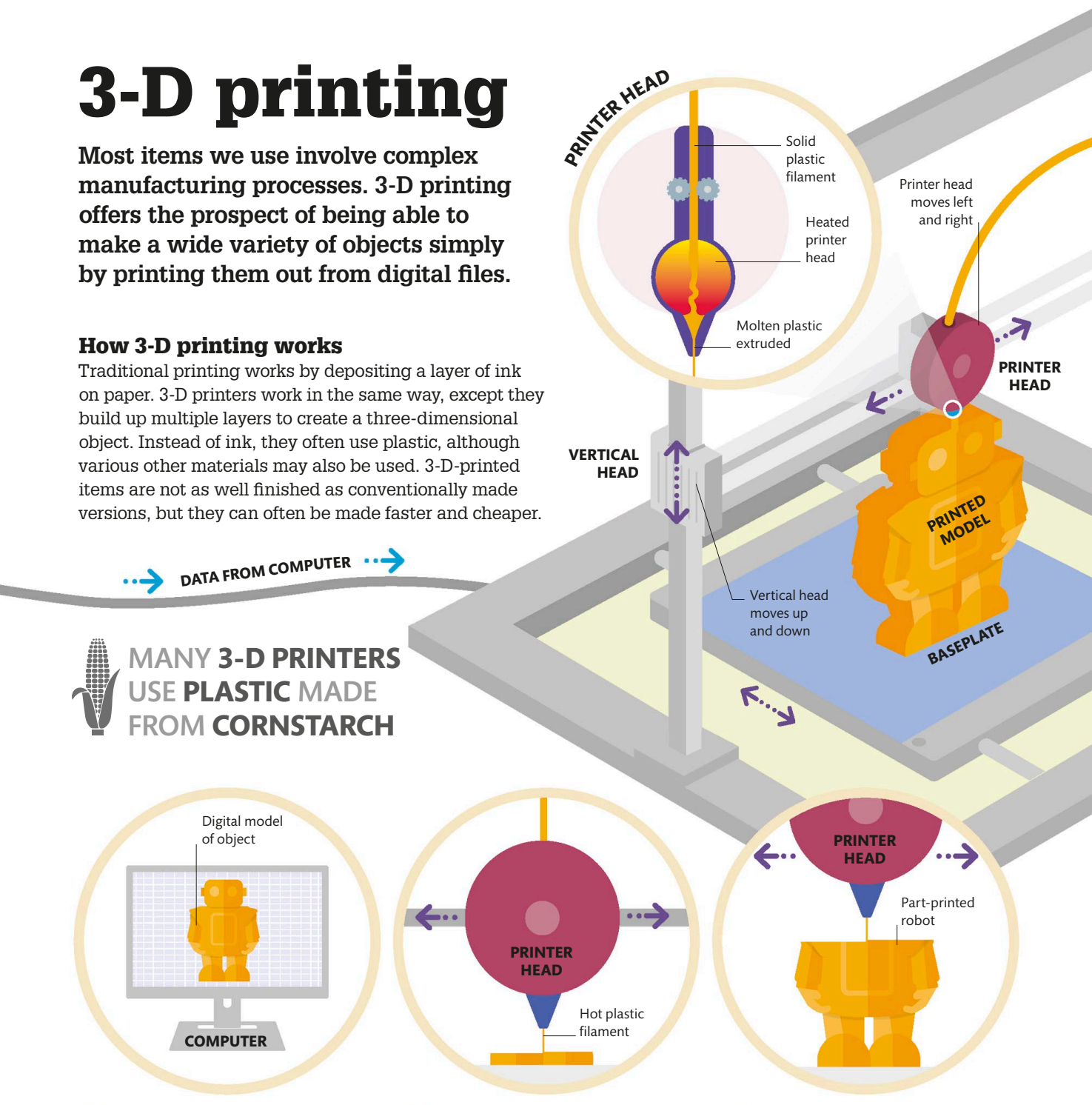
3-D printing

Most items we use involve complex manufacturing processes. 3-D printing offers the prospect of being able to make a wide variety of objects simply by printing them out from digital files.

How 3-D printing works

Traditional printing works by depositing a layer of ink on paper. 3-D printers work in the same way, except they build up multiple layers to create a three-dimensional object. Instead of ink, they often use plastic, although various other materials may also be used. 3-D-printed items are not as well finished as conventionally made versions, but they can often be made faster and cheaper.



MANY **3-D PRINTERS** USE **PLASTIC** MADE FROM **CORNSTARCH**

1 Computer design
3-D printing starts with the creation of a 3-D digital model in a computer. The model may be produced by special software or by scanning an object with a laser and then digitizing and processing the scanning data.

2 Start printing
A plastic filament is fed through the printer head, which contains a heating element that melts the plastic. Data from the computer moves the printer head from side to side, the vertical head up and down, and the baseplate back and forth.

3 Building up layers
The printed object is gradually built up layer by layer from the bottom upward. As each layer is added, the molten plastic cools and solidifies. Depending on the size and complexity of the object, it may take up to several hours to print.